MILITARY SPECIFICATION

RADAR SET, AN/APS-88A

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be an airborne search radar and shall provide means for searching for and detecting surface targets.

1.2 Classification - The AN/APS-88A Radar Set shall consist of the following items:

Items	Type Designation	Max. Allow. Size (Inches) W	H	L	Max. Allow. Wt. Lbs.	Appl. Para.
Receiver-Transmitter, Radar	RT-497B/APS-88	19-3/16	21-½ diam.		90.0	3.4.1
Antenna	AS-1179/APS-88A	42	39	42	61.0	3.4.2
Amplifier, Electronic Control	AM-2086/APS-88	5-½	6-11/16	10-5/8	7.2	3.4.3
Gyroscope Displacement	CN-790/APS-88A	6-3/4	6-5/8	8-3/8	6.7	3.4.4
Indicator, Azimuth Range	IP-483B/APS-88	9-11/16	8-5/8	26-½	20	3.4.5
Control Antenna	C-3825/APS-88A	5-3/4	6-3/8	6-1/8	2.8	3.4.6
Control, Radar Set	C-2719/APS-88	5-3/4	7	5-3/16	3.0	3.4.7
Synchronizer, Electrical	SN-230/APS-88	5-9/32	7-5/8	11-3/4	10.0	3.4.8
Switch, Waveguide	SA-511/U	3-½	5-9/16	4-¼	3.5	3.4.9

/FSC 5841/

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Items	Type Designation	Max. Allow. Size (Inches) W	H	L	Max. Allow. Wt. Lbs.	Appl. Para.
Dummy Load, Electrical	DA-146/U	2-3/4	2-3/4	5-11/16	1.2	3.4.10
Fan, Vaneaxial (2)	HD-395/APS-88	5-7/8	3-½	7-3/8	1.5 ea.	3.4.11
Compressor, Rotary, Power Driven	HD-485/APS-88A	6-3/8	9-9/16	14	9.5	3.4.12
Mounting	MS-91404-S1				1.2	3.4.8.2
Mounting	MT-2278/APS-88				1.0	3.4.3.4
Mounting	MT-2101/APS-88				2.7	3.4.5.2

1.3 Associated Equipment - The Radar Set AN/APS-88A shall operate with the following associated equipment which shall not be supplied as a part of this equipment:

Radio Receiving Set	AN/ARR-58
Computer Group; Navigational	AN/ASA-13A
Computer-Display Set, Navigational	AN/ASN-30
Computer Set, Navigational	AN/AYK-2

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply:

SPECIFICATIONS

Military

MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for

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MIL-T-5422	Testing, Environmental Aircraft Electronic Equipment
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack - or Console-Mounted
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Deliver of
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment

STANDARDS

Military

MIL-STD-242 (NAVY)	Military Standard Equipment Parts
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MS-25245	Connections, Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, or 115/200-Volt 3-Phase AC, or Combinations
MS-28053	Meter, Time Totalizing, 400-Cycle

PUBLICATION

Bureau of Naval Weapons

SAR-300	Material Changes and Material Bulletins, Preparation of

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2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code DCS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

3.2.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 225 pounds.

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3.2.2 Operating Life -

3.2.2.1 Operational Stability - The equipment shall operate with optimum performance for 50 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.2.2 Reliable Operating Life - The equipment shall have a reliable operating life of at least 300 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.2.3 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.2.4 Time Totalizing Meter - A time totalizing meter conforming with MS-28053 shall be mounted externally on the Receiver-Transmitter, Radar RT-497B/APS-88 to indicate the total hours the transmitter high voltage has been energized.

3.2.3 Cabling and Connections -

3.2.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

3.2.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

3.2.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

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3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient ($30^{\circ} \pm 10^{\circ}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0V AC and/or 27.5 ± 0.5V DC

3.2.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 1 equipment, except as modified herein.

3.2.10.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400.

3.2.10.2 Altitude - The complete equipment shall operate satisfactorily at barometric pressures ranging from 30 inches Hg down to 8.9 inches Hg (sea level to 30,000 ft. altitude) with pressurization, or at pressures down to 23.5 inches Hg (7,000 ft. altitude) without pressurization.

3.2.11 Warm-Up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 15 minutes at extreme service conditions.

3.2.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) AC Power (Three-Phase) 115/200V, Category "B" 1700VA

(2) DC Power 28V Category "B" 10 Amps under standard conditions and 10.5 Amps at extreme environmental conditions of temperature and altitude.

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3.2.12.1 Input Power Connections - Input power connections shall be in accordance with MS-25245.

3.2.12.2 Voltage Protection - The equipment shall not be damaged by voltages below the minimum voltages specified herein and shall automatically resume normal operation when the voltage returns within the specified limits.

3.2.12.3 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.13 Standby Provisions - The equipment shall provide for a standby position whereby rated voltage shall be applied to all tube filaments. All low voltages shall be applied, also.

3.2.13.1 Magnetron Heater Schedule - The equipment shall provide for the automatic application of the correct heater voltage for each operating function. Voltage values shall be as recommended by the magnetron manufacturer for this specific application.

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.3.1 Operation - The equipment shall operate as a complete airborne pulsed radar system. The primary function of the system is to transmit pulses of radio frequency energy and receive, demodulate, and display the resulting echo signals on a PPI to provide range and bearing information on targets-of-interest to an operator. The radar shall also be compatible with an IFF system as specified herein.

3.3.2 Frequency Characteristics - The radar shall be capable of operating throughout a frequency range of 8500 to 9600 MC.

3.3.3 Modulation Characteristics - The transmitter shall be pulse modulated at repetition rates of 2,000, 1,025, and 200 pulses per second by pulse widths of 0.35, 0.80, and 4.5 microseconds, respectively.

3.3.4 Transmitter Power Output - The equipment shall deliver a peak radio frequency power of not less than 45KW and not more than 125KW. Power output shall be measured colorimetrically at the output flange of the RT-497B/APS-88.

3.3.5 System Noise Figure - The overall system noise figure with IN23WE crystals shall not exceed 9.3 db over a tuning range of 8500 to 9600 mcs.

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3.3.6 Anti-Jam Provisions - The following anti-jam provisions shall be incorporated in the receiving system:

(1) Sensitivity Time Control

(2) COAC Controlled IF Strip

3.3.7 Indication - Visual range and azimuth indication shall be provided on a 7-inch Cathode Ray Tube (CRT).

3.3.8 IFF Provisions -

3.3.8.1 IFF Radar Interconnection - In order to interconnect IFF and radar equipments, to permit the IFF equipment to transmit challenge signals in synchronization with the radar transmission, and to permit presentation of the replies received by the IFF equipment along with radar echoes on the radar indicator, the following provisions shall be made:

3.3.8.2 Synchronization - Synchronization of the accessory IFF equipment is to be accomplished by a positive pulse from the receiver-transmitter.

3.3.8.3 Display - Provisions shall be made to display IFF video on the IP-483B/APS-88 radar indicator along with the radar echoes.

3.3.9 System Pressure - All units in the system requiring pressurization shall be part of a common pressurized system. All components of the pressurized system shall be capable of safely withstanding 30 psig. With the Compressor, Rotary, Power Driven HD-485/APS-88A in the system but not operating, and with the remainder of the equipment operating normally, the maximum rate of pressure loss shall not exceed 6 pounds in 4 hours when starting from an initial pressure of 15 psig. This leakage rate shall be based on the standard condition as specified in paragraph 3.2.9.

3.4 Detail Requirements -

3.4.1 Receiver-Transmitter, Radar RT-497B/APS-88 - The receiver-transmitter shall meet the following requirements.

3.4.1.1 Function - This unit shall generate radio frequency pulses and deliver them to the antenna unit and shall receive radio frequency pulses from the antenna, convert these pulses to an intermediate frequency, amplify and detect them and furnish a video output to the indicator unit.

3.4.1.2 Form Factor and Mounting - The general form factor shall be that of a cylinder. The unit shall be capable of being completely contained within a cylinder 21-1/2 inches in diameter by 19-3/16 inches high. The 21-1/2 inch diameter shall include shock mounts, connectors and fuse box. The center section shall be a casting. The casting shall support the unit components and external connections shall be made through the casting. There shall be pressure seal covers on each side. The diameter of the center section, measured across

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the cover retaining clamps, shall not exceed 16 inches. The 19-3/16 inches height is without external cooling fans attached. The unit shall be provided with three shock mounts for installation in the aircraft, so positioned that the longitudinal axis of the cylinder shall be vertical.

3.4.1.3 Weight - The weight of this unit excluding shock mounts shall not exceed 90 pounds.

3.4.1.4 Cooling - The receiver-transmitter unit shall be cooled by both external and internal blowers. The magnetron tube shall be cooled by a blower. The external blowers shall be Fan, Vaneaxial HD-395/APS-88. The covers for the receiver-transmitter shall contain mounting provisions for the external blowers.

3.4.1.5 Pressurization - This unit shall be a part of the pressurized system and shall conform to the requirements of 3.3.9.

3.4.1.6 Power Capacity and VSWR - The radio frequency elements of the transmitter shall be capable of handling peaks of at least 125 kw of radio frequency power in the specified frequency range under all operating conditions. The VSWR of the receiver-transmitter duplexer shall not exceed 1.3:1 in the transmit direction. The VSWR limit for the complete waveguide assembly shall be 1.5:1 in the transmit direction. This measurement shall include the flexible connection to the magnetron, a dummy load isolator, the duplexer, the directional coupler, and the bulkhead output flange. Also in this measurement a dummy TR tube shall be used to simulate the TR tube in the ionized condition. The VSWR for the complete waveguide assembly shall not exceed 1.7:1 in the receive direction.

3.4.1.7 Load Isolation - A load isolator shall be provided between the magnetron and duplexer. The minimum isolation shall be 10 db. Maximum insertion loss shall be 1 db and maximum VSWR 1.1 to 1.

3.4.1.8 Frequency - The transmitting and receiving frequencies shall be tunable over a range of 8500 to 9600 mcs.

3.4.1.9 Power Output - The transmitter shall deliver a peak radio frequency power of not less than 45 kw and not more than 125 kw. Power output shall be measured calormetrically at the output flange of the RT-497B/APS-88.

3.4.1.10 Pulse Repetition Frequency - The radio frequency output pulses shall have the following repetition frequencies within ± 10%:

Range (Nautical Miles)	Search	Warning
3-30	2000 Pulses/Sec	200 Pulses/Sec
50	1025	200
100	200	200
240	200	200

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3.4.1.11 Pulse Lengths - The output pulse shall have the following lengths in microseconds within ±10%, unless otherwise noted:

Range (Nautical Miles)	Search	Warning
3-30	0.35 ± 20%	4.5
50	0.80	4.5
100	4.5	4.5
240	4.5	4.5

Pulse length shall be measured at the 90% amplitude point on the detected radio frequency output pulse.

3.4.1.12 Pulse Rise Time - The rise time of the radio frequency pulses shall not exceed 0.10 usec between 20 and 80% amplitude levels.

3.4.1.13 Pulse Delay - The 10% of maximum point on the rise of the radio frequency output pulse shall occur not later than 0.5 usec after the 10% of maximum point on the rise of the input modulating pulse.

3.4.1.14 Crystal Protection - A crystal shutter shall not be used. The waveguide switch shall perform this function when installed.

3.4.1.15 Crystal Current - Means shall be provided to permit measuring of search crystal current from the radar at the receiver-transmitter unit. The crystal current shall not exceed 1.0 milliampere.

3.4.1.16 Overall Recoevery Time - When equipment is stabilized at an external ambient temperature no lower than 20°C, the receiver shall return within 3 db of full sensitivity in less than 6 usec after completion of the transmitted pulse.

3.4.1.17 Automatic Frequency Control - The operation of the AFC circuits shall be such that with any available repetition rate the receiver shall automatically maintain optimum conditions of reception for radar operation without necessity for manual tuning when the receiver-transmitter is coupled to and operated with the specified antenna.

3.4.1.18 Manual Tuning Provisions - It shall be possible by means of an external switch located on Control, Radar Set C-2719/APS-88 to disable the automatic frequency control of the search local oscillator. In conjunction with the AFC Manual switch there shall be a knob control which shall tune the local oscillator.

3.4.1.19 Receiver Section -

3.4.1.19.1 Function - The receiver unit shall amplify the IF signals from the mixer detect, and by means of a cathode follower, provide video signals of the proper polarity and amplitudes to the indicating system.

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3.4.1.19.2 IF Amplifier Characteristics -

(1) IF shall be 30 mc/s.

(2) IF bandwidth shall be 1.5 ± 0.3 mc/s.

(3) IF gain shall be sufficient to provide at least 1.0 volt of noise at the second detector, and shall be adjustable.

(4) The characteristic response of the receiver shall be COAC having a pedestal compression range of at least 80 db, and shall be adjustable. The recovery time following an 80 db signal shall be 2 usec or less. The receiver shall have a relay activated by a switch on Control, Radar Set C-2719/APS-88 to change the receiver from COAC to linear operation.

(5) The IF noise figure shall not exceed 1.3 db.

3.4.1.20 System Noise Figure - The system noise figure, when measured with a standard microwave noise source at the antenna connection to the receiver-transmitter, shall not exceed 9.3 db over the tuning range of 8500 to 9600 mcs.

3.4.1.21 Directional Coupler - The directional coupler shall be connected to the waveguide to provide a means for removing or inserting small amounts of radio frequency power. It shall be located inside the receiver-transmitter housing, with test connections accessible from outside the receiver-transmitter.

3.4.1.21.1 Attenuation - The one-way loss for transmitted power shall not exceed 0.4 db at any frequency in the specified range.

3.4.1.21.2 Standing Wave Ratio - The voltage standing wave ratio between 8500 and 9600 mcs shall not exceed 1.2:1 (with matched loading).

3.4.1.21.3 Electrical Characteristics - The directional coupler shall be of the bi-directional type, capable of coupling out transmitted and reflected radio frequency power.

3.4.1.21.4 Coupling - The coupling of the transmitted power from main line into the auxiliary line of the directional coupler shall be between the limits of 28 and 36 db.

3.4.1.21.5 Coupling Magnitude - The magnitude of the coupling for a particular coupler within an accuracy of ± 0.3 db, shall be marked on the outside of the receiver-transmitter unit near the connector used for test purposes. The coupling shall be indicated for at least three frequencies, one each, at low, high and mid-frequency in the 8500 to 9600 megacycle band.

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3.4.1.21.6 Directivity - The directivity of the directional coupler shall be better than 15 db.

3.4.1.22 DC Power Supply - A power supply shall be included in the receiver-transmitter unit which shall supply regulated dc voltages for the radar set.

3.4.1.22.1 Voltages - The dc voltages available from this unit shall be 330 unregulated and +300, +250, +150, -105, and -255 volts regulated. It shall be possible to adjust the +300, +250, +150, and -105 voltages.

3.4.1.22.2 Regulation - The dc voltage regulation shall be in accordance with the following:

Voltage	Regulation Tolerance
+330 V Unregulated	± 20%
+300 V Regulated	± 1%
+250 V Regulated	± 1%
+150 V Regulated	± 1%
-105 V Regulated	± 1%
-255 V Regulated	-5% + 3%

3.4.1.23 Transmitter Section - The transmitter shall be manually tunable and shall deliver radio frequency power at any frequency in the range 8500 to 9600 mcs. The output power shall not be less than 45 kw peak and shall be delivered under any of the operating conditions specified in paragraphs 3.4.1.10 and 3.4.1.11.

3.4.1.24 Modulator Section - The high voltage rectifiers, charging diode, and inverse diode shall be of the single module semiconductor type.

3.4.1.24.1 Pulse Forming Network - The pulse forming network shall provide means to disable the high voltage when the internal line pressure reaches 20 ± 2 psi gauge.

3.4.2 Antenna AS-1179/APS-88A - The antenna shall meet the following requirements.

3.4.2.1 Function - The antenna system shall provide for focusing, radiating, and receiving radio frequency energy in the frequency range 8500 to 9600 mcs as well as for varying the special orientation of the radar beam and relaying to the indicator the azimuth bearing of such orientation. The azimuth information shall be suitable for generating a PPI input.

3.4.2.2 Stabilization and Tilt - Synchro type information provided by the Gyroscope, Displacement CN-790/APS-88A shall cause the antenna

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reflector to be automatically stabilized to within ± 1° of level position with respect to the horizontal. This stabilization accuracy shall be maintained up to 25° ± 1° on the roll axis and 15° ± 1° on the pitch axis. Gyroscope inputs up to ± 60° shall not cause damage to the antenna or stabilization equipment, and the system shall operate normally when the error returns to within the specified roll and pitch limits.(The time duration for the specified abnormal attitude requirement shall not exceed 5 minutes). Means shall be provided for the operator to bring the antenna to a neutral position by operation of the "STAB-OUT" switch on Control, Antenna C-3825/APS-88A. Manual operation of the tilt controls on Control, Antenna C-3825/APS-88A shall cause the beam to tilt with respect to the horizontal between the limits of 10 ± 1° elevation and 15 ± 1° depression.

3.4.2.3 Form Factor and Mounting - The overall dimensions shall not exceed 34-¼ inches in height, 42 inches in width, 42 inches in length, excluding the gyro, with zero tilt and centered stabilization. The reflector shall be 42 inches in width and mounted eccentric to the azimuth axis so that when rotated through any position the tips shall be within a 24-inch arc, whose center lies on the azimuth axis. With the reflector in the maximum pitch, roll and tilt attitude, the maximum overall height shall not exceed 39 inches. Three vibration isolators shall be provided for installation in the aircraft. The antenna shall provide 360° rotation and shall fit into the space provided in an aircraft for Antenna AS-567/APS-33B and shall utilize the same radome.

3.4.2.4 Weight - The weight of this unit shall not exceed 61 pounds.

3.4.2.5 Rotating Provisions -

3.4.2.5.1 Azimuth Scan Rate - The reflector assembly shall be capable of rotating continuously about its vertical axis at a rate of 6 ± 1 rpm or 28 ± 8 rpm. The "SCAN" mode of operation will be controlled by the Control, Antenna C-3825/APS-88A.

3.4.2.5.2 Sector Scan - The reflector shall be capable of azimuth sector scanning at a rate of 36° per second. The sector width shall be adjustable from 40° to 140° by means of the "SECTOR WIDTH" control on Control, Antenna C-3825/APS-88A. The center of the sector shall be adjustable through 360° from the Control, Antenna C-3825/APS-88A. In addition provision for a fixed sector width, adjustable in 30° increments up to 240°, shall be available for nose-mounted antenna installations (3.4.6.5.(2)).

3.4.2.5.3 Searchlighting Provision - The antenna reflector shall be capable of being searchlighted, i.e., "stopped", at any position in response to operator control. It shall be possible to stop the antenna at any azimuth angle within ± 0.5°.

3.4.2.5.4 Braking - The reflector shall be braked or clamped by a suitable braking action whenever the radar set is in the "OFF" position.

3.4.2.6 Data Input and Take-Offs -

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3.4.2.6.1 Relative Bearing - Synchro signals shall be provided so that the azmuthal relationship between the antenna reflector and the indicator sweep can be maintained at zero phase. When the radiation from the antenna is in the direction of the aircraft heading the radial sweep on the indicator shall be in a vertical position extending from the center to the top edge of the cathode ray tube.

3.4.2.6.2 True Bearing - Provision shall be made for the antenna to be true north or south stabilized by signals derived from the aircraft compass. In the true north mode the radiation from the antenna shall be in the direction of true north when the radial sweep on the indicator is in a vertical position extending from the center to the top edge of the cathode ray tube. In the south mode, the display shall be rotated 180°.

3.4.2.6.3 Synchros - A total of three (3) synchros shall follow the azimuth position of the antenna. One shall provide azimuth position information to Azimuth and Range Indicator IP-483B/APS-88. Two additional synchros shall be provided for auxiliary equipment and their shafts shall rotate with the antenna within 0.1°.

3.4.2.6.4 Heading Marker - A heading marker signal shall be generated each time the reflector dish is directed along the centerline of the aircraft forward of the radar installation. This signal shall be provided to the indicator unit.

3.4.2.7 RF Waveguide and Radiation Elements -

3.4.2.7.1 Power Capacity - The radio frequency elements shall be capable of handling peaks of at least 125 kw of radio frequency power in the specified frequency range under any combination of operating service conditions.

3.4.2.7.2 Standing Wave Ratio - The voltage standing wave ratio in the antenna shall not exceed 1.7:1 when measured at frequencies of 8500, 9000, 9375 and 9600 mcs when the antenna is radiating into free space and set at any tilt angle within the limits specified in paragraph 3.4.2.2.

3.4.2.7.3 Size of Waveguide - The waveguide shall mate with a waveguide whose inside dimensions are 0.900 by 0.400 inches.

3.4.2.7.4 Input Waveguide - The radio frequency input connections to the antenna system shall be a choke flange.

3.4.2.7.5 Pressurization - The waveguide shall be a part of the pressurized system.

3.4.2.8 Reflector - The reflector shall be a nominal 42 by 20 inches in size, having the curvature of a paraboloidal surface.

3.4.2.8.1 Polarization - The radiated electric field shall be horizontally polarized when the antenna is mounted normally in an erect attitude.

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3.4.2.8.2 Spoiler Plate - A spoiler plate designed for the cosecant squared patterns specified in paragraphs 3.4.2.8.6 and 3.4.2.8.7 shall be provided. The plate shall be easily detached and stowed by means of fasteners.

3.4.2.8.3 Pencil Beam - Horizontal Pattern - The 3 db beamwidth measured at any tilt angle within the specified range shall not exceed 2.6 degrees and the side lobe level shall be down at least 22 db from maximum power without the spoiler plate attached when measured at a frequency of 9450 mcs. When measured at 8500 mcs and 9600 mcs the beamwidth shall not exceed 2.8 degrees and the side lobe level shall be down at least 21 db from maximum power.

3.4.2.8.4 Pencil Beam - Vertical Pattern - The 3 db beam width measured at any tilt angle within the specified range shall not exceed 5 degrees and the side lobe level shall be at least 20 db down from maximum power without the spoiler plate attached when measured at a frequency of 9450 mcs. When measured at 8500 mcs and 9600 mcs the beamwidth shall not exceed 5.2 degrees and the side lobe level shall be at least 19 db down from maximum power.

3.4.2.8.5 Pencil Beam - Antenna Gain - The antenna gain shall be at least 33.5 db at any tilt angle within the specified range without the spoiler plate attached when measured at a frequency of 9450 mcs. When measured at frequencies of 8500 and 9600 mcs the antenna gain shall be at least 32.5 db. The direction of maximum power for any given azimuth setting shall occur at that setting within 0.4 degrees for depression or elevation angles of up to 25 degrees.

3.4.2.8.6 CSC^2 Beam-Horizontal Pattern - The 3 db beamwidth measured at any tilt angle within the specified range shall not exceed 3.2 degrees with the spoiler plate attached when measured at frequencies of 8500, 9450, and 9600 mcs.

3.4.2.8.7 CSC^2 Beam - Vertical Pattern - With the spoiler plate attached the antenna pattern shall conform to the law, $I = KCSC^2\theta_0$ from $\theta = \theta_0$ to $\theta = \theta_1$ with $\pm$ 5.0 db between depression angles of $\theta_0 = 1.5^\circ$ and $\theta_1 = 18^\circ$. This requirement shall be met at frequencies of 8500, 9450, and 9600 mcs. The side lobe level shall be at least 20 db down from maximum power when measured at 9450 mcs. When measured at frequencies of 8500 and 9600 mcs the side lobe level shall be at least 19 db down from maximum power.

3.4.2.8.8 CSC^2 Beam - Antenna Gain - The antenna gain shall be at least 31.5 db at any tilt angle within the specified range with the spoiler plate attached when measured at a frequency of 9450 mcs. When measured at frequencies of 8500 and 9600 mcs the antenna gain shall be at least 31.0 db.

3.4.3 Amplifier, Electronic Control AM-2086/APS-88 - The amplifier, electronic control shall meet the following requirements:

3.4.3.1 Function - This unit shall contain all servo amplifiers, relays, etc., to stabilize and control the position of the antenna reflector except the azimuth drive servo system, which shall be in the antenna unit.

3.4.3.2 Size - The size of this unit shall not exceed 6-11/16

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inches height by 8-½ inches length by 5-½ inches width, exclusive of mounting base, handles, and connectors. The connectors and handle shall project from the front panel not more than 2-1/8 inches.

3.4.3.3 Weight - The weight of this unit shall not exceed 7.2 pounds, excluding the mounting base.

3.4.3.4 Mounting - The unit shall be mounted on a vibration isolation type shock mounting MT-2278/APS-88.

3.4.4 Gyroscope, Displacement CN-790/APS-88A - The gyroscope, displacement shall meet the following requirements:

3.4.4.1 Function - This gyroscope shall provide the sensing voltage which in connection with the servo amplifiers in the electronic control amplifier unit shall cause the antenna dish to remain in a stable plane within the limits specified in 3.4.2.2.

3.4.4.2 Mounting - The unit shall be mounted in accordance with standard practices for aircraft mounted gyroscope. It shall be possible to mount this unit at any point within 85 feet of the antenna unit or integrally on the antenna base, as desired.

3.4.4.3 Size and Weight - The external dimensions of this unit shall not exceed those of a cylinder, 8-3/8 inches in diameter and 7 inches in height. The weight of this unit shall not exceed 6.7 lbs.

3.4.5 Indicator, Azimuth and Range IP-483B/APS-88 - The indicator shall meet the following requirements:

3.4.5.1 Function - The indicator unit shall present a visual indication of the demodulated echo signals, beacon signals, and internal reference marks, which shall form a plan position indicator display.

3.4.5.2 Form Factor and Mounting - The basic dimensions of this unit shall be 9-½ inches wide, 8-5/8 inches high and 13 inches deep. Protruding from the rear approximately 1 inch shall be the electrical connectors. Approximately 1-5/8 inches projection shall be allowed from the front panel for the knobs and the C.R.T. bezel. Located horizontally on the front panel shall be a single handle, extending approximately 2-1/8 inches. A detachable viewing hood shall extend approximately 12-5/8 inches from the front panel

3.4.5.2.1 Mounting MT - 2101/APS-88 - The IP-483B/APS-88 shall be capable of mounting on vibration isolator base, Mounting MT-2101/APS-88, conforming to the applicable requirements of Specification MIL-C-172. The vibration isolator base shall have the centerline of the front two isolators located 4-1/8 inches from the metal front panel of the basic indicator unit. The centerline of the two rear isolators shall be located 7-7/8 inches from the centerline of the two forward mounts. The isolators shall be spaced equally about the longitudinal axis and shall have their centerlines spaced 6-½ inches apart. The overall dimensions of the isolator base shall be: width, 10-5/16 inches; the length, 14-3/16 inches with hinged retaining screws adding

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1-7/8 inches, height from mounting surface to indicator mounting surface approximately 1-3/4 inches. Quick disconnect type fasteners, not requiring tools, shall be used to secure the unit to its mounting. The controls shall be operable from the front panel. The unit shall perform as specified when installed with its longitudinal axis horizontal, or vertical or inclined at any angle between these limits. The overall dimensions, including the viewing hood but excluding the mounting base, shall not exceed; 9-11/16 inches wide, 8-5/8 inches high and 26-½ inches long. The weight of this mount shall not exceed 2.7 pounds.

3.4.5.3 Weight - The weight of the IP-483B/APS-88 excluding the mounting shall not exceed 20 pounds.

3.4.5.4 Construction -

3.4.5.4.1 Shielding - Adequate shielding shall be provided of metal having sufficiently high shielding properties to protect the indicator and surrounding instruments from electrostatic and magnetic fields.

3.4.5.4.2 Receptacles - Receptacles for interconnecting cables shall be mounted in the back of the unit.

3.4.5.5 Cathode Ray Tube - The cathode ray tube shall be a 7 inch type 7AQP7 and shall be replaceable from the front by removal of the cursor and filter assembly and the indicator dust cover.

3.4.5.6 Plug-In Assemblies - The various circuits incorporated in the indicator unit shall be plug-in assemblies and shall utilize sub-miniature type vacuum tubes or transistors when practical. All sub-units shall utilize Winchester connectors or an approved equivalent. Each unit shall be adequately identified in accordance with Specification MIL-N-18307.

3.4.5.6.1 Electronic Control Amplifier - This amplifier shall accept error signals from the control transformer in the indicator unit and shall furnish an output voltage to drive the motor which turns the cathode ray tube deflection coil in synchronism with the antenna.

3.4.5.6.2 Power Supply - The high voltage power supply shall be rated at 10 kv ± 10% at 100 micro-amps load.

3.4.5.6.3 Video Amplifier - This amplifier shall receive low level video signal voltages from the FTC and clamp unit, sonobuoy receiver, marker generator, and IFF sources, mix, gate and amplify these signals and supply them to control-electrodes on the cathode ray tube of the indicator.

3.4.5.6.4 FTC and Clamp - This unit shall receive and amplify the low level video signals from the receiver-transmitter and supply the amplified signals to the video amplifier unit. When the FTC is switched in the signals shall be differentiated in a short time constant circuit. This unit shall contain the clamp circuitry for the heading marker circuits.

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3.4.5.6.5 CRT Protection - A circuit shall be provided to prevent the face of the CRT from being burned when the radar is switched off.

3.4.5.7 Cursor and Filter - It shall be possible by means of an adjustable cursor to measure azimuth angles with an accuracy of $\pm 1^{o}$. The cursor line and protractor markings shall be made visible by means of edge mounted lights of adjustable brilliancy. The cursor movement shall be independent of the filter. It shall be possible to readily insert or delete grease pencil data on the filter.

3.4.5.8 Range Marker Lights - Range marker lights shall be provided around the upper rim of the tube and shall operate so as to indicate automatically the position of the range selector switch and the interval in nautical miles between range marks. These range marker lights shall be visible from inside the visor.

3.4.5.9 Visor - A visor shall be provided to shield the face of the indicator from external light. This visor shall be quickly mountable and demountable from the unit.

3.4.5.10 Type of Indication - The indication shall consist of radial range sweeps originating at the center of the usable cathode ray tube face, the angular positions of which shall indicate the relative azimuth position of the reflector portions of the antenna system. Indication shall be provided on the face of a 7-inch tube. The rotation of the sweep line shall be clockwise when the antenna is rotating clockwise and counterclockwise when the antenna is rotating counterclockwise.

3.4.5.11 Accuracy of Presentation -

3.4.5.11.1 Azimuth Accuracy - The accuracy of the azimuth follow-up shall be such that the azimuth position of the indicator servo shall follow the angular rotation of the antenna system to within 1.5^{o} on continuous rotation.

3.4.5.11.2 Sweep Jitter - The jitter of the origin of any sweep shall not exceed 3/64 inches.

3.4.5.11.3 Sweep Linearity - The spacing between any two range marks excluding the first range mark shall not differ from the average spacing by more than 10%. The variable ranges shall be excluded from these requirements.

3.4.5.12 Visual Characteristics -

3.4.5.12.1 Gain - With the intensity adjusted so that the sweep trace is barely visible in dark surrounding with no video input, the video gain shall be sufficient for 50 millivolt applied video signals to be visually detected on the face of the cathode ray tube.

3.4.5.12.2 Focus - The indication shall be in focus at all points of the usable tube face with one proper adjustment of the focusing control.

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The control shall be a maintenance adjustment located internal to the unit for electrostatic focus tubes.

3.4.5.12.3 Aircraft Heading Marker - A radial line shall be displayed on the face of the indicator tube which shall indicate the actual heading of the aircraft.

3.4.5.12.4 Range Marks - It shall be possible to display and vary the intensity of range marks on the face of the indicator tube. Intervals shall be as specified in paragraphs 3.4.8.9 and 3.4.8.10

3.4.5.13 Video Bandwidth - The video response from the video input receptacle to the video connector at the cathode ray tube shall be flat within 3 decibels from 300 cps to 4 mcs.

3.4.5.14 Controls - The requirements of Specification MIL-P-7788 relative to panel lighting are applicable to the control panel of the indicator unit. Radioactive luminescent materials shall not be used. The following controls shall be placed on the indicator unit; and shall be so located that they can easily be grasped by the operator. The knob form (for controls No. 1 through No. 10 listed below) shall be in accordance with MIL-STD-242C (NAVY) Section 700.8 dated 15 August 1958 only as indicated below.

(1) Dual Light Dimmer to control azimuth ring illumination - Triangular (Form E; color, gray)

(2) Variable Range Control (3 to 30 nautical miles) - Cross (Form F; color, gray)

(3) Indicator Tube Intensity Control - Triangular (similar to Form E; color, gray)

(4) Range Marker Intensity and ON-OFF Switch - Triangular with Skirt and Lighted Position Indicator (similar to Form E; color, gray)

(5) Heading Marker Amplitude - Triangular (Form E; color, gray)

(6) Sonobuoy A Video Gain - Circular (similar to Form C; color, gray)

(7) Sonobuoy B Video Gain - Circular (similar to Form C; color, gray). This control shall be concentric with Control (6)

(8) N-S Relative Bearing Switch - Hexagonal with Lighted Position Indicator (similar to Form A; color, gray)

(9) FTC Switch - (MIL-STD-242C (NAVY), Section 700.8 not applicable)

(10) Radar Video Gain - Circular with Lighted Position Indicator (similar to Form C; color, gray)

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3.4.6 Control, Antenna C-3825/APS-88A - The antenna control shall be of the console, rack mounted type in accordance with Specification MIL-C-6781 and shall meet the following requirements:

3.4.6.1 Function - The control unit shall provide manual controls by means of which the operating condition of the antenna may be selected. Controlled functions shall include the tilt angle of the radar beam, antenna scan, stabilization, and heading.

3.4.6.2 Form Factor and Mounting - The dimensions and mounting of the unit shall be in accordance with Specification MIL-C-6781, except that it shall be of the following overall size. Front panel dimensions; 5-3/4 inches wide, 6-3/8 inches high. A cover shall extend from the back of the panel and shall be the following size; 5 inches wide, 6-¼ inches high and 4 inches deep. An electrical connector shall extend through the cover approximately 1 inch. The total depth from the mounting surface including the connector shall be approximately 5 inches. The edge-lighted plastic panel and the knobs shall extend from the front panel approximately 1-3/16 inches. Captive fasteners shall be in accordance with Specification MIL-F-25173, for mounting in accordance with Specification MIL-C-6781. The maximum outside dimensions of the unit shall not exceed; 5-3/4 inches wide, 6-1/8 inches deep and 6-3/8 inches high.

3.4.6.3 Contents - This unit shall contain the controls described in 3.4.6.5, and shall have an edge-lighted panel. This panel shall conform to Specification MIL-P-7788. The identifying words "RADAR" shall be arranged in a vertical column at the edge of this panel.

3.4.6.4 Weight - The weight of this unit shall not exceed 2.8 pounds.

3.4.6.5 Controls - The following operating controls shall be provided on the front panel of this unit:

(1) "TILT" - This knob control shall permit adjustment of the mechanical tilt angle of the radar antenna reflector so that the radar beam will tilt with respect to its horizontal reference axis from 15° ± 1° down, to 10° ± 1° up.

(2) "SCAN" - This multiple position knob control shall be provided to control the condition of antenna rotation. There shall be four positions "STOP", "SECTOR", "FULL-6", and "FULL-28". When placed in the "STOP" position, this control shall stop the azimuth rotation of the antenna at the setting of the "ANT HEADING" control. In the "SECTOR" position, the antenna shall scan the sector selected by the operation of the "SECTOR WIDTH" and "ANT HEADING" controls. In the "FULL-6" or "FULL-28"

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positions, the antenna shall normally scan 360°, at the RPM selected, with PPI presentation. If the antenna is connected for a nose mounted installation, the FULL positions of this control will provide a sector scan centered ahead of the aircraft. The sector scan provided in the "FULL" position is changed by maintenance adjustment only, in 30° increments, and the speed remains 6 RPM. All other control positions remain unchanged.

(3) "ANT HEADING" - This knob control shall permit positioning or "searchlighting" of the antenna by the operator when the scan switch is in the "STOP" position. It shall permit selection of the azimuth angle about which the antenna will sector scan when SCAN switch is in "SECTOR".

(4) "STAB-OUT" - This two-position toggle switch shall be provided to select the stabilization of the antenna reflector. When in the "STAB" or up position, this control shall cause the antenna reflector to be automatically stabilized. When in the "OUT" or down position, the reflector vertical axis shall be locked perpendicular to the aircraft deck plane throughout rotation.

(5) "SECTOR WIDTH" - This knob control shall permit the selection of the sector width from 40° to 140°.

3.4.7 Control, Radar Set C-2719/APS-88 - The radar set control shall be of the rack mounted console type in accordance with Specification MIL-C-6781 and shall meet the following requirements:

3.4.7.1 Function - This unit shall provide controls for the operation of the radar set.

3.4.7.2 Form Factor and Mounting - The dimensions and mounting of the unit shall be in accordance with Specification MIL-C-6781, except it shall be the following overall size. Front panel dimensions, 5-3/4 inches wide, 7 inches high. A cover shall extend from the back of the panel and be the following size: 5 inches wide, 6-7/8 inches high and 3 inches deep. An electrical connector shall extend through the cover approximately 1 inch. The total depth from the mounting surface including the connector shall be approximately 4 inches. The edge-lighted plastic panel and knobs shall extend from the front panel approximately 1-3/16 inches. Captive fasteners shall be in accordance with Specification MIL-F-25173, for mounting in accordance with Specification MIL-C-6781. The maximum outside dimensions of the unit shall not exceed; 5-3/4 inches wide, 5-3/16 inches deep and 7 inches high.

3.4.7.3 Contents - This unit shall contain three lights; one green for indicating when the high voltage is on, another amber for indicating

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when the radar is on standby; i.e., high voltage off, and the third amber for indicating when dummy load is on. It shall contain all controls necessary for operating the transmitting and receiving functions of the radar set. The controls which pertain to the operation of the receiver shall be grouped together and those which pertain to the operation of the transmitter shall be grouped together. An edge-lighted plastic panel conforming to Specification MIL-P-7788 shall cover the front surface of the control. The identifying word "RADAR" shall be arranged in a vertical column at the left edge of this panel. The dummy load light shall be provided with a shutter type dimmer.

3.4.7.4 Weight - The weight of this unit shall not exceed 3.0 pounds.

3.4.7.5 Controls - The following operating controls shall be provided on the front panel of this unit and shall be labeled as follows:

(1) "FUNCTION" - This knob control shall energize the system and select the system function. Positions shall be "OFF", "STDBY", "SEARCH" and "WARNING".

(2) "ANT-DUMMY LOAD" - This double throw toggle switch shall provide for normal operation of the radar antenna when in "ANT" position and shall switch the radio frequency output of the transmitter into the dummy load when in the "DUMMY LOAD" position. Switching shall be so arranged that with the set "OFF", the dummy load position is automatically attained.

(3) "RANGE" - This knob control shall permit selection of range for PPI display.

(4) "AFC-MAN" - This double-throw switch shall establish automatic frequency control of the local oscillator when in the "AFC" position. When in the "MAN" position, this switch shall permit adjustment of the search local oscillator by means of the "MAN-TUNE" control.

(5) "MAN-TUNE" - This control shall provide manual frequency control by varying the value of repeller voltage of the search local oscillator over a range of 40 volts, provided the "AFC-MAN" switch is in the "MAN" position.

(6) "GAIN" - This knob control shall permit the setting of a proper signal and noise level at the receiver output, and shall provide for a gain reduction of at least 70 db. A switch to permit linear or COAC operation shall be a part of the gain control.

(7) "STC" - These two knob controls shall be mounted on

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concentric shafts and shall provide for adjustment of the sensitivity time control (STC). The lower knob (outer shaft) shall vary the exponential decay time of the (STC) bias voltage between zero and 600 microseconds. The upper knob shall provide for the selection of the gain reduction characteristic of the (STC) voltage.

(8) "TEST METER" - This meter and selector switch shall be provided to monitor magnetron current, receiver 1 and 2 and AFC 1 and 2 crystal current.

3.4.8 Synchronizer, Electrical SN-230/APS-88 - The Synchronizer shall meet the following requirements:

3.4.8.1 Function - This synchronizer shall furnish basic timing and synchronization signals to the modulator and to other auxiliary equipment.

3.4.8.2 Form Factor and Mounting - The dimensions of the unit shall be in conformance with MS91402-S1 case, length 9-7/8 inches, height 7-5/8 inches, width 5-1/8 inches. The MS standard 3/16 inch projection, front and rear, for attachment to an MS vibration isolator, shall be in addition to the above dimensions. All controls, electrical connectors, and a single handle shall be accommodated on the front panel. Maximum projection of these items shall be 1-3/4 inches. The unit shall be mounted on a vibration isolation type shock mounting MS-91404-S1 conforming to Specification MIL-C-172. The overall dimensions of the unit, excluding the mounting base, shall not exceed; 5-9/32 inches wide, 7-5/8 inches high and 11-3/4 inches long.

3.4.8.3 Weight - This unit shall not weigh more than 10.0 pounds, excluding the mounting. The weight of the mounting should not exceed 1.2 pounds.

3.4.8.4 Contents - This unit shall contain range sweep, range markers and synchronizing signal generators.

3.4.8.5 Electrical Connections - Electrical connections shall be provided as necessary to connect this unit to associated circuits in accordance with the requirements of this specification.

3.4.8.6 Panel Controls - No operating controls shall be included on this unit.

3.4.8.7 Electrical Outputs - This synchronizer shall provide the following outputs; Modulator trigger, Indicator Gate, Sweep and range mark signals.

3.4.8.8 Plug-In Assemblies - The various circuits incorporated in the synchronizer unit shall be plug-in assemblies and shall utilize subminiature type vacuum tubes when practical. Each unit shall be adequately identified in accordance with Specification MIL-N-18307.

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3.4.8.8.1 Gate and Sweep Generator - The gate and sweep generator shall accept a trigger from the synchronizing signal generator and shall generate the following:

(1) Range gate and sweep voltages of proper length as determined by the control box setting.

(2) An unblanking gate for the video amplifier.

(3) A sweep voltage output for the sweep amplifier.

3.4.8.8.2 Range Mark Generator - The range mark generator shall accept the gate from the gate and sweep generator and shall generate range marks as indicated in 3.4.8.10 which shall be fed to the video amplifier.

3.4.8.8.3 Synchronizing Signal Generator - This unit shall provide basic timing to the sweep and marker circuits during standby and to the transmitter during operation.

3.4.8.8.4 Sweep Amplifier - The sweep amplifier shall accept the voltage from the gate and sweep generator and provide the current control for the deflection yoke in Indicator, Azimuth and Range IP-483B/APS-88.

3.4.8.9 Range Sweeps -

3.4.8.9.1 Ranges - The ranges to be provided by this unit shall be as follows on search operation:

(1) 3 to 30 nautical miles (continuously variable)

(2) 50 nautical miles (fixed)

(3) 100 nautical miles (fixed)

(4) 240 nautical miles (fixed)

3.4.8.9.2 Types of Sweep Ranges - Slant ranges only shall be provided by this equipment.

3.4.8.9.3 Timing - The range sweep shall start within 0.5 microseconds after the 90% amplitude point of the leading edge of the transmitted radar pulse as delivered to the indicator from the receiver.

3.4.8.10 Range Markers - Range marker pulses shall be generated in intervals of time equivalent to the following ranges within 1% accuracy.

(1) 1 or 5 nautical miles (selectable by operator) for 3 to 30 mile ranges.

(2) 10 nautical miles for 50 mile range.

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(3) 20 nautical miles for 100 mile range.

(4) 30 nautical miles for 240 mile range.

The zero marker shall occur within one microsecond after the 90% of maximum amplitude points of the leading edge of the transmitted pulse, as delivered to the indicator from the receiver, or within a time corresponding to 1% of the sweep in use, whichever is greater. The range markers shall be displayed in the indicator only when desired by the operator.

3.4.8.10.1 Marker Widths - The persistent traces caused by the markers on the indicator tube face shall not exceed 0.5 millimeter for any normal settings on the controls. All reference markers shall have approximately the same brilliance for any adjustment of the range mark intensity. The brilliance of the range marks shall not vary appreciably when switching from one range to another.

3.4.9 Switch, Waveguide SA-511/U -

3.4.9.1 Function - The waveguide switch shall be capable of switching the radio frequency output of the transmitter to Dummy Load, Electrical DA-146/U when the switch on Control, Radar Set C-2719/APS-88 is placed in the "Dummy Load" position.

3.4.9.2 Dimensions - The external dimensions of this unit shall not exceed those of a cylinder 3-½ inches in diameter and 5-9/16 inches in height, except for an electrical connector. The connector shall not add more than 3/4 inch to the 3-½ inch dimension.

3.4.9.3 Weight - The weight of this unit shall not exceed 3.5 pounds.

3.4.9.4 VSWR - The VSWR shall not exceed 1.12 over the frequency range of 8500 to 9600 mcs in either switched condition.

3.4.9.5 Electrical Connections - The following electrical connections shall be provided:

(1) Waveguide to antenna

(2) Waveguide to Dummy Load

(3) Control Wire - +28 volts

(4) Input Waveguide

3.4.9.6 Isolation - With the radar radio frequency output switched to the dummy load, the isolation between the radio frequency input and the antenna connection shall be at least 50 db.

3.4.10 Dummy Load, Electrical DA-146/U -

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3.4.10.1 Function - The dummy load shall be capable of providing a termination for the transmitter when the radio frequency channel is switched to the dummy load position.

3.4.10.2 Dimensions - The overall dimensions shall not exceed 2-3/4 inches in height by 2-3/4 inches in width by 5-11/16 inches in length.

3.4.10.3 Weight - The weight of this unit shall not exceed 1.2 pounds.

3.4.10.4 Power - The dummy load shall be capable of absorbing 275 kw peak power and 250 watts average power.

3.4.10.5 VSWR - The VSWR shall not exceed 1.12 over the frequency range of 8500 to 9600 mcs.

3.4.10.6 Electrical Connections - The dummy load shall be connected to the transmitter side of the radio frequency output by means of the dummy load termination of the Switch, Waveguide SA-511/U.

3.4.11 Fan, Vaneaxial, HD-395/APS-88 - Two fans shall be provided with each equipment to meet the requirements specified below.

3.4.11.1 Function - The fans shall cause a flow of air sufficient for external cooling of Receiver-Transmitter, Radar RT-497B/APS-88.

3.4.11.2 Dimensions - The general form factor shall be that of a cylinder 5-7/8 inches in diameter, with a small cylindrical projection oneone axis resulting in an overall height of 3-7/16 inches. A rectangular projection on one side shall extend the diameter in one direction to an overall of 7-3/8 inches. The height of this projection shall not exceed 2-5/8 inches. The overall dimensions of the unit shall not exceed; 5-7/8 inches wide, 3-$\frac{1}{2}$ inches high and 7-3/8 inches deep.

3.4.11.3 Weight - The weight shall not exceed 1.5 pounds each.

3.4.11.4 Mounting Provisions - The units shall be capable of being rigidly mounted, by means of four captive screws, to the Receiver-Transmitter end covers, so that air is forced through the cover ducts.

3.4.12 Compressor, Rotary, Power Driven HD-485/APS-88A - One pressurizing unit shall be furnished to meet the following requirements:

3.4.12.1 Function - This unit shall be used to maintain the pressure in the pressurized system at 20 $\pm$ 2 psia up to an altitude of 30,000 feet and incorporate a pressure relief valve set to relieve at 24 $\pm$ 2 psia.

3.4.12.2 Dimensions - Overall dimensions shall not exceed 6-3/8 inches by 9-9/16 inches by 14 inches.

3.4.12.3 Weight - The weight of this unit shall not exceed 9.5 pounds.

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3.4.12.4 Operation - The pressurizing system shall dehydrate the pressurized air and shall be of such a design that no small particles of dehydrator agent can pass into the pressurized system.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating conditions. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

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4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.2.5 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by this government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper

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corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours

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without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6 hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6 hour period specified above may be composed of two 3 hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

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(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing reliability test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows; except that when a Production Sample is requested and delivered to a government laboratory for test (See paragraph 4.3) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity).

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

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4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests).

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Tests
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)

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D. C. Voltage 27.0 ± 2.0 volts

4.5.2 Test Periods - This test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number.

(2) The circuit reference symbol number.

(3) The part function.

(4) Name of the manufacturer.

(5) Nature of the failure.

(6) The number of hours which the part operated prior to failure.

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

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4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use -

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 $\pm$ 0.5 V DC and 115 $\pm$ 1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance of reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings

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should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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